

Problem 1

(a) Let V be a vector space over $\mathbb{R}$ of dimension $\dim V < \infty$, and let $V \ni x \mapsto \|x\|$, $V \ni x \mapsto \|x\|_1$ be norms on V . Show that the norms $\|\cdot\|$, $\|\cdot\|_1$ are equivalent on V , in the sense that there exist constants $C_1 > 0$, $C_2 > 0$ such that for all $x \in V$,

$$C_1\|x\| \leq \|x\|_1 \leq C_2\|x\|.$$

Hint: You may assume that $V = \mathbb{R}^n$ and that one of the two norms is Euclidean.

(b) Show that for each $d \in \mathbb{N}$ there exists $C_d > 0$ such that for all polynomials p of degree $\leq d$,

$$\sup_{[0,1]} |p(x)| \leq C_d \int_0^1 |p(x)| dx.$$

Solution. (a): We will first show that any two norms on $\mathbb{R}^n$ are equivalent. It suffices to show equivalence between any norm $\|\cdot\|_1$ and $\|\cdot\|$, the Euclidean Norm. Suppose $\|\cdot\|_1 \sim \|\cdot\| \sim \|\cdot\|_2$, then $\forall x \in \mathbb{R}^n$ we have

$$C_1\|x\| \leq \|x\|_1 \leq C_2\|x\| \Rightarrow \|x\| \leq \frac{1}{C_1}\|x\|_1 \text{ and } \frac{1}{C_2}\|x\|_1 \leq \|x\|$$

and

$$D_1\|x\| \leq \|x\|_2 \leq D_2\|x\|.$$

Thus

$$\frac{D_1}{C_2}\|x\|_1 \leq D_1\|x\| \leq \|x\|_2 \leq D_2\|x\| \leq \frac{D_2}{C_1}\|x\|_1.$$

Thus $\|x\|_1 \sim \|x\|_2$. Thus we will show equivalence between any norm $\|\cdot\|_1$ and $\|\cdot\|$. Furthermore suppose we have that for all $x \in S^{n-1} \subset \mathbb{R}^n$ that

$$C_1 = C_1\|x\| \leq \|x\|_1 \leq C_2\|x\| = C_2$$

. Then for all $y \in \mathbb{R}^n$:

$$C_1\|y\| = C_1\|y\| \left\| \frac{y}{\|y\|} \right\|_1 \leq \|y\| \left\| \frac{y}{\|y\|} \right\|_1 = \|y\|_1.$$

Similarly

$$\|y\|_1 = \|y\| \left\| \frac{y}{\|y\|} \right\|_1 \leq C_2\|y\| \left\| \frac{y}{\|y\|} \right\|_1 = C_2\|y\|.$$

Thus it suffices to show equivalence on strictly on S^{n-1} .

$\|\cdot\|_1$ Continuous with respect to $\|\cdot\|$: Let

$$M = \left(\sum_{i=1}^n \|e_i\|_1^2 \right)^{\frac{1}{2}}$$

$\forall \varepsilon > 0$, let $\delta = \frac{\varepsilon}{M}$ then for $\|x - y\| < \delta$:

$$\begin{aligned} \|\|y\|_1 - \|x\|_1\| &= \|\|y - x + x\|_1 - \|x\|_1\| \\ &= \|\|y - x\|_1\| \\ &= \|y - x\|_1 \end{aligned}$$

Let

$$y - x = \sum_{i=1}^n a_i e_i$$

. Thus

$$\begin{aligned} \|y - x\|_1 &\leq \sum_{i=1}^n \|a_i e_i\|_1 \\ &\leq \sum_{i=1}^n |a_i| \|e_i\|_1 \\ &\leq \left(\sum_{i=1}^n |a_i|^2 \right)^{\frac{1}{2}} \left(\sum_{i=1}^n \|e_i\|_1^2 \right)^{\frac{1}{2}} = M \|x - y\| < \varepsilon. \end{aligned}$$

S^{n-1} compact in $\mathbb{R}^n$. Thus $\|\cdot\|_1$ achieves minimum and maximum on S^1 . Thus $\exists C_1, C_2 > 0$ such that

$$C_1 \leq \|x\|_1 \leq C_2$$

for $x \in S^{n-1}$. Thus all norms on $\mathbb{R}^n$ are equivalent. Let V be a finite dimensional vector space with norms $\|\cdot\|_a$ and $\|\cdot\|_b$. There exists some $T : \mathbb{R}^n \rightarrow V$ a linear isomorphism. Define induced norms on $\mathbb{R}^n$ where

$$\begin{aligned} \|x\|_A &:= \|Tx\|_a \\ \|x\|_B &= \|Tx\|_b. \end{aligned}$$

Checking the norm axioms (it suffices to check one):

$$\begin{aligned} \|x + y\|_A &= \|Tx + Ty\|_a \leq \|Tx\|_a + \|Ty\|_a = \|x\|_A + \|y\|_A \\ \|\lambda x\|_A &= \|T\lambda x\|_a = |\lambda| \|Tx\|_a = |\lambda| \|x\|_A \\ 0 &= \|x\|_a = \|Tx\|_A \Leftrightarrow Tx = 0 \Leftrightarrow x = 0 \end{aligned}$$

Thus $\|\cdot\|_A \sim \|\cdot\|_B$ in $\mathbb{R}^n$ and $\exists C_1, C_2 < \infty$ where

$$C_1 \|x\|_A \leq \|x\|_B \leq C_2 \|x\|_A.$$

$\forall v \in V$, let $x = T^{-1}v$.

$$C_1 \|v\|_a = C_1 \|x\|_A \leq \|x\|_B = \|v\|_b \leq C_2 \|x\|_A = C_2 \|v\|_a.$$

Thus $\|\cdot\|_a \sim \|\cdot\|_b$.

(b): Let V be the vector space of polynomials with real coefficients of degree at most d . We have the following vector space isomorphisms:

$$V = \frac{\mathbb{R}[x]}{(x^{d+1})} \cong \mathbb{R}\langle 1, x, x^2, \dots, x^d \rangle \cong \mathbb{R}^{d+1}.$$

$\|\cdot\|_\infty$ and $\|\cdot\|_1$ are valid norms on V . Since V is a finite dimensional vector space, by part a, there exists a C_2 such that

$$\|p\|_\infty \leq C_1 \|p\|_1.$$

Problem 2 : Basic Exam, Spring 2017

For $n \geq 1$, let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that

$$|f_n(x)| \leq 1 + \frac{n}{1 + n^2 x^2}, \quad x \in [0, 1],$$

and define

$$F_n(x) = \int_0^x f_n(t) dt, \quad x \in [0, 1].$$

Show that the sequence $(F_n)_{n \geq 1}$ admits a subsequence that converges pointwise on $[0, 1]$.

Solution.

$$\begin{aligned} |F_n(x)| &= \left| \int_0^x f_n(t) dt \right| \\ &\leq \int_0^x |f_n(t)| dt \\ &\leq \int_0^x \left(1 + \frac{n}{1 + n^2 t^2} \right) dt \\ &= \int_0^{nx} \left(\frac{1}{n} + \frac{1}{1 + u^2} \right) du \end{aligned}$$

Solving the integral we see:

$$|F_n(x)| \leq x + \arctan(nx) \leq 1 + \frac{\pi}{2} = M$$

Thus $(F_n)_{n=1}^\infty$ is bounded. By the first part of Arzela Ascoli Theorem (diagonalization argument), F_n admits a subsequence which converges pointwise on $\mathbb{Q} \cap [0, 1]$, let $G_k = F_{n_k}$ denote such a subsequence. $\forall k \in \mathbb{N}$ for $x, y \in [0, 1]$:

$$\begin{aligned}
|G_k(x) - G_k(y)| &= \left| \int_y^x f_{n_k}(t) dt \right| \\
&\leq \int_y^x |f_{n_k}(t)| dt \\
&\leq \int_y^x \left(1 + \frac{n_k}{1 + n_k^2 t^2} \right) dt \\
&= \left[\frac{1}{n_k} u + \arctan(u) \right]_{n_k y}^{n_k x} \\
&= (x - y) + \arctan(n_k x) - \arctan(n_k y)
\end{aligned}$$

Note that

$$\lim_{k \rightarrow \infty} (\arctan(n_k x) - \arctan(n_k y)) = 0$$

$\forall x \in [0, 1], \forall \varepsilon > 0$, by density of $\mathbb{Q} \cap [0, 1]$ in $[0, 1]$, there exists a $q \in \mathbb{Q} \cap [0, 1]$ such that $0 < x - q < \frac{\varepsilon}{6}$. There also exists a N_1 such that $\forall k \geq N_1$:

$$\arctan(n_k x) - \arctan(n_k q) < \frac{\varepsilon}{6}.$$

Furthermore there exists a $N_2 \in \mathbb{N}$ such that

$$\forall a, b \geq N_2$$

: $|G_a(q) - G_b(q)| < \frac{\varepsilon}{3}$. Let $N = \max(N_1, N_2)$. Thus $\forall a, b \geq N$ we have that

$$\begin{aligned}
|G_a(x) - G_b(x)| &\leq |G_a(x) - G_a(q)| + |G_a(q) - G_b(q)| + |G_b(q) - G_b(x)| \\
&< \frac{\varepsilon}{6} + \frac{\varepsilon}{6} + \frac{\varepsilon}{3} + \frac{\varepsilon}{6} + \frac{\varepsilon}{6} = \varepsilon
\end{aligned}$$

Thus $(G_k(x))_{k=1}^{\infty}$ is Cauchy and thus G_k converges pointwise to some function

$$G(x) = \lim_{k \rightarrow \infty} G_k(x)$$

Problem 3

Let f be a continuous function on the compact interval $\{x \in \mathbb{R}; a \leq x \leq b\}$ such that the right-hand derivative $f'_r(x)$ exists when $a \leq x < b$. Assume that $f'_r(x) \geq C$ for all $x \in [a, b)$. Show that $f(b) - f(a) \geq C(b - a)$.

Hint: Use the continuity method.

Solution. $\forall \varepsilon > 0$, let $E_\varepsilon = \{x \in [a, b] : f(y) - f(a) \geq (C - \varepsilon)(y - a), \forall y \in [a, x]\}$. Note if $x \in E_\varepsilon \Leftrightarrow \forall y \in [a, x]$ we have that $[a, y] \in E_\varepsilon$.

1. Nonempty: $a \in E$.
2. Closed: Let (x_n) be a sequence in $E \subset [a, b]$ such that $x_n \rightarrow L \in [a, b]$. Thus

$$f(x_n) - f(a) \geq (C - \varepsilon)(x_n - a).$$

Taking the limit on both sides, we see $f(L) - f(a) \geq (C - \varepsilon)(L - a)$. It suffices to assume that $\forall n \in \mathbb{N}, x_n < L$ otherwise $L \in E_\varepsilon$ automatically. $\forall x \in [a, L), \exists N \in \mathbb{N}$ such that $L - x_n < L - x \Rightarrow x_n > x$. Since $[a, x_n] \subset E_\varepsilon \Rightarrow [a, x] \subset E_\varepsilon$. Thus $[a, L] \subset E_\varepsilon$.

3. Open: $\forall t_0 \in E$. It suffices to assume that $t_0 \neq b$, otherwise we have nothing to prove. We have that

$$(f(t_0) - f(a)) > (C - \varepsilon)(t_0 - a).$$

Existence of $f_r'(t_0)$ and fact that $f_r(t_0) \geq C$: $\exists \delta > 0$ such that $\forall h \in (0, \delta)$:

$$\frac{f(t_0 + h) - f(t_0)}{h} \geq C - \varepsilon \Rightarrow f(t_0 + h) - f(t_0) \geq (C - \varepsilon)h$$

Thus

$$f(t_0 + h) - f(a) \geq (C - \varepsilon)(t_0 + h - a).$$

Furthermore, since the statement is true for all $x \in [0, t_0]$ and that for $h' \in (0, h) \subset (0, \delta)$ the statement is true by the above for $t_0 + h'$. We have that the statement is true for all $[a, t_0 + h]$ for $h \in (0, \delta)$.

Thus if $t_0 \in E_\varepsilon$ then $(\max(a, t_0 - \delta), t_0 + \delta) \subseteq E_\varepsilon$. Thus E_ε open.

By hence E_ε is open and closed and hence it is $[a, b]$, by connectedness. Since $\forall \varepsilon, f(b) - f(a) \geq (C - \varepsilon)(b - a) \Rightarrow f(b) - f(a) \geq C(b - a)$.

Problem 4

Let (X, d) be a metric space and let $E \subseteq X$. We say that a function $f : E \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant $L > 0$ (L -Lipschitz for short) if

$$|f(x) - f(y)| \leq Ld(x, y), \quad \forall x, y \in E.$$

- (a) Let $(f_\alpha)_{\alpha \in J}$ be a collection of L -Lipschitz functions $f_\alpha : E \rightarrow \mathbb{R}$. Show that the functions

$$E \ni x \mapsto \inf_{\alpha \in J} f_\alpha(x), \quad E \ni x \mapsto \sup_{\alpha \in J} f_\alpha(x)$$

are L -Lipschitz on E , if finite at one point.

- (b) Let $f : E \rightarrow \mathbb{R}$ be L -Lipschitz. Show that the function

$$F(x) = \inf_{y \in E} (f(y) + Ld(x, y)), \quad x \in X,$$

is an L -Lipschitz extension of f . Show also that F is the largest L -Lipschitz extension of f , in the sense that if $G : X \rightarrow \mathbb{R}$ is any L -Lipschitz extension of f then $G \leq F$. Can you find the smallest L -Lipschitz extension of f ?

Solution. (a): Let $g : E \rightarrow \mathbb{R} \cup \{-\infty\}$ and $h : E \rightarrow \mathbb{R} \cup \{\infty\}$ where

$$g(x) = \inf_{\alpha \in J} f_\alpha(x)$$

$$h(x) = \sup_{\alpha \in J} f_\alpha(x).$$

Suppose there exists x_0 such that $g(x_0) > -\infty$. $\forall x \in E$ and $\forall \alpha \in J$:

$$|f_\alpha(x) - f_\alpha(x_0)| \leq Ld(x, x_0) \Rightarrow -Ld(x, x_0) \leq f_\alpha(x) - f_\alpha(x_0) \leq Ld(x, x_0).$$

Thus

$$f_\alpha(x_0) - Ld(x, x_0) \leq f_\alpha(x) \leq Ld(x, x_0) + f_\alpha(x_0).$$

Since $g(x_0) \leq f_\alpha(x_0)$, we have that $\forall \alpha$,

$$g(x_0) - Ld(x, x_0) \leq f_\alpha(x_0) \Rightarrow g(x_0) - Ld(x, x_0) \leq g(x).$$

Thus $-Ld(x, x_0) \leq g(x) - g(x_0)$. $g(x) \leq f_\alpha(x) \leq f_\alpha(x_0) + Ld(x, x_0)$. Since $f_\alpha(x) - Ld(x, x_0) \leq f_\alpha(x_0) \Rightarrow g(x) \leq f_\alpha(x) \leq g(x_0) + Ld(x_0, x)$. thus

$$|g(x) - g(x_0)| \leq Ld(x, x_0).$$

Note that thus $\forall x \in E$, $g(x)$ is finite. The proof is symmetric for h .

(b): Let $\{g_y(x) := f(y) + Ld(x, y)\}_{y \in E}$ be a collection of functions. $\forall x, z \in X$:

$$\begin{aligned} |g_y(x) - g_y(z)| &= L|d(x, y) - d(y, z)| \\ &\leq L|d(x, z) + d(z, y) - d(y, z)| \\ &= Ld(x, z). \end{aligned}$$

Thus $g_y(x)$ is L -Lipschitz. We need to show that F is finite for some $x \in X$. Fix $y_0 \in E$:

$$F(y_0) = \inf_{y \in E} (f(y) + Ld(y, y_0)).$$

By the fact that f is L Lipschitz:

$$f(y_0) - f(y) \leq Ld(y, y_0) \Rightarrow f(y_0) \leq f(y) + Ld(y, y_0) \Rightarrow f(y_0) \leq F(y_0).$$

Thus $F(y_0) > -\infty$. Thus F is L Lipschitz by part a.

Suppose $G : X \rightarrow \mathbb{R}$ be a L -Lipschitz extension of f . Fix a $x \in X$. $\forall y \in E$ we have

$$G(x) - G(y) = G(x) - f(y) \leq Ld(x, y) \Rightarrow G(x) \leq f(y) + Ld(x, y).$$

Thus $\forall y \in E$: $G(x) \leq f(y) + Ld(x, y)$. Thus

$$G(x) \leq \inf_{y \in E} (f(y) + Ld(x, y)) = F(x) \Rightarrow G \leq F.$$

The smallest L -Lipschitz extension of f is

$$H(x) = \sup_{y \in E} (f(y) - Ld(x, y)).$$

Problem 5

Let $f : [0, \infty) \rightarrow \mathbb{R}$ be continuous and assume that for each $x \geq 0$ the sequence $f(nx)$ tends to 0 as $n \rightarrow \infty$. Show that $f(t) \rightarrow 0$ as $t \rightarrow \infty$.

Hint: Given $\varepsilon > 0$, consider

$$F_n = \{x \geq 0 : \forall p \geq n, |f(px)| \leq \varepsilon\}, \quad n = 1, 2, \dots$$

Solution. For $p \in \mathbb{N}$, let $g_p : [0, \infty) \rightarrow \mathbb{R}$ where $g_p(x) = f(px)$. For a fixed $\varepsilon > 0$, consider

$$\begin{aligned} F_n &:= \{x \geq 0 : \forall p \geq n, |f(px)| \leq \varepsilon\} = \{x \geq 0 : \forall p \geq n, -\varepsilon \leq f(px) \leq \varepsilon\} \\ &= \bigcap_{p \geq n} g_p^{-1}([-\varepsilon, \varepsilon]) \end{aligned}$$

$g_p^{-1}([-\varepsilon, \varepsilon])$ is closed, which implies that F_n is closed since it's an intersection of closed sets. Note $0 \in F_1$. Note that $F_i \subseteq F_{i+1}$ and

$$\bigcup_{n \in \mathbb{N}} F_n = [0, \infty).$$

Take the contrapositive of the Baire Category Theorem, thus $\exists N \in \mathbb{N}$ with nonempty interior. Thus $\forall n \geq N, \exists [a, b] \in F_n$. Thus $\forall x \in F_N, |g_N(x)| = |f(Nx)| < \varepsilon$. Thus $\forall n \geq N$ and $\forall y \in [na, nb] \Rightarrow \frac{y}{n} \in [a, b]$, we have that $|f(y)| \leq \varepsilon$. Thus

$$\text{if } y \in \bigcup_{i=1}^{\infty} [(N+i)a, (N+i)b] \Rightarrow |f(y)| < \varepsilon.$$

Let $I_n = [(N+n)a, (N+n)b]$. Let $h : \mathbb{N} \rightarrow \mathbb{R}$ be the function which tells the gap between I_{i+1} and I_i .

$$h(n) := (N+n+1)a - (N+n)b.$$

If $h(n) > 0$: then there is no overlap between I_{n+1} and I_n . If $h(n) \leq 0$, then there is overlap between I_{n+1} and I_n .

$$\begin{aligned} h(n)(N+n+1)a - (N+n)b &= na - nb + (N+1)a - Nb \leq 0 \\ -n(b-a) &\leq a - N(b-a) \Rightarrow n \geq N - \frac{a}{b-a} \end{aligned}$$

Thus for all $n \geq \lceil N - \frac{a}{b-a} \rceil = M$, $I_n \cap I_{n+1} \neq \emptyset$. Furthermore note $|I_n| = n(b-a)$, thus as $n \rightarrow \infty, |I_n| \rightarrow \infty$. Thus you stitch together the intervals and you see $y \in \bigcup_{n \geq M} I_n = [M(a), \infty) \Rightarrow \|f(y)\| < \varepsilon$. Thus $f \rightarrow 0$ as $t \rightarrow \infty$.

Problem 6 : Basic Exam, Spring 2017

Let (X, d) be a bounded metric space and let $C(X)$ denote the space of bounded continuous real-valued functions on X endowed with the supremum norm. Suppose $C(X)$ is separable.

- Show that for every $\varepsilon > 0$ there exists a countable set $Z_\varepsilon \subseteq X$ such that

$$\forall x \in X \quad \exists z \in Z_\varepsilon \quad \text{such that} \quad \forall y \in X, \quad |d(x, y) - d(z, y)| < \varepsilon.$$

- Deduce that X is separable.

Solution. $\forall x \in X$, let $d_x := X \rightarrow \mathbb{R}$ where $d_x(y) = d(x, y)$. By definition of X , $\exists M > 0$ such that $\forall (y, z) \in X^2$, $d(y, z) \leq M \Rightarrow d_x \leq M$. $\forall \varepsilon > 0$ and $\forall y \in X$, let $\delta = \varepsilon$ then for all $y_1 \in B_\delta(y)$ we have

$$|d_x(y) - d_x(y_1)| = |d(x, y) - d(x, y_1)| \leq d(y, y_1) < \delta \leq \varepsilon.$$

Thus $d_x \in C(X)$. Let $S = \{d_x : x \in X\}$. S subspace of $C(X)$ and hence separable. Thus there exists a countable dense set D of S . By definition of S ,

$$D = \{d_x : x \in A\}$$

for a countable subset $A \subset X$. Thus $\forall \varepsilon > 0$ and $\forall x \in X$, there exists $z \in A$ such that

$$\|d_x - d_z\|_\infty < \varepsilon.$$

Thus $A = Z_\varepsilon$ for all $\varepsilon > 0$.

$\forall x \in X$, and $\forall \delta > 0$ by the previous problem $\exists a \in A$ such that for all $y \in X$:

$$|d(x, y) - d(a, y)| < \delta.$$

Let $y = x$:

$$|d(x, x) - d(a, x)| = d(a, x) < \delta \Rightarrow a \in A \cap B_\delta(x).$$

Thus A is a countable dense subset of X and hence X is separable.